

Everything you need to know about Toric Geometry

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1

Toric Variety

Toric varieties are varieties that can be parameterized by monomials. These naturally form all linear subspaces, half-spaces, fans, and rational cones, but also form shapes like the cubic curve $t \mapsto (t^2, t^3)$. These form a large set of examples that are worth getting comfortable with.

1.1 Definition and basic Properties

Geometrically, the torus is usually represented as $(S^1)^n$, the n -product of circles S^1 . Its fundamental group is $\mathbb{Z}^n$.

Definition 1.1.1: [Algebraic] Torus

Let $(\mathbb{C}^*)^n$ be the affine variety which also has the component-wise multiplication structure. Then this is the *canonical* or *standard torus*.

Any affine variety T that is a [regular and group] isomorphism to $(\mathbb{C}^*)^n$ is called a *torus*. In this setting $\mathbb{C}^*$ is denoted $\mathbb{G}_m$ or $\mathbf{G}_m$.

We shall call morphisms that are both regular and group morphisms *toric morphisms*.

The group structure of $(\mathbb{C}^*)^n$ is given by pointwise multiplication:

$$(t_1, \dots, t_n) \cdot (s_1, \dots, s_n) = (t_1 s_1, \dots, t_n s_n)$$

Note that the standard torus is homotopic to $(S^1)^n$, motivating the terminology. By sticking to $(\mathbb{C}^*)^n$, we may take it as the maximal Spectrum:

$$(\mathbb{C}^*)^n = \text{mSpec}(\mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}])$$

Let $\text{Hom}_{\text{GVar}}(T, \mathbb{C}^*)$ be the collection of morphism from T to $\mathbb{C}^*$ (namely characters) where the maps are variety and group morphisms. We can find elements of this set for each $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$, where we get a character $\chi^m : T \rightarrow \mathbb{C}^*$ given by:

$$\chi^m(t_1, \dots, t_n) = t_1^{m_1} \cdots t_n^{m_n} \quad m \in \mathbb{Z}^n$$

In fact, *all* characters are of this form (determine all the group homomorphisms, and the eliminate the conjugates as they are not variety morphisms)¹. This complete description gives rise to a simple lattice:

Lemma 1.1.2: Character Lattice

Let $T \cong (\mathbb{C}^*)^n$ be a torus. Then the torus induces a lattice $M \cong \mathbb{Z}^n$.

Proof :

The isomorphism bijectively maps the hom-sets.

The dual perspective is to take all one-parameter subgroups of $(\mathbb{C}^*)^n$ (i.e. all morphisms from $\mathbb{C}^*$ to T). These all arise as variety+group morphisms $\lambda : \mathbb{C}^* \rightarrow T$, and if $T = (\mathbb{C}^*)^n$, then they are all given by a $u = (b_1, \dots, b_n) \in \mathbb{Z}^n$:

$$\lambda^u(t) = (t^{b_1}, \dots, t^{b_n})$$

Or similarly, we may take

$$\text{Hom}_{\text{GVar}}(M, \mathbb{Z}) \cong \mathbb{Z}^n \quad (1.1)$$

The proof that it is a lattice should be clear, and hence we give it a name:

Definition 1.1.3: Character Lattice

Let T_N be a affine toric variety. Then the *character lattice* or *colattice* is the lattice formed by it's characters. Each elements of $\text{Hom}_{\text{GVar}}(T, \mathbb{C}^*)$ will be represented as χ^m .

Note that the one-parameter groups preserve more information than the image. For example, if $u = (1, 0, \dots, 0)$ and $v = (2, 0, \dots, 0)$, then $\text{im}(\lambda^u) = \text{im}(\lambda^v)$, however the two are considered to give different one-parameter subgroups.

From 1.1, we see that the one-parameter subgroup from a free abelian group N of rank equal to the dimension of N . With these two maps, given a choice of basis $T \cong (\mathbb{C}^*)^n$ we have the following natural map:

$$\langle -, - \rangle : M \times N \rightarrow \mathbb{Z} \quad \chi^m \circ \lambda^u : \mathbb{C}^* \rightarrow \mathbb{C}^* \in \text{Hom}(\mathbb{C}^*, \mathbb{C}^*) \cong \mathbb{C}^*$$

In other words, their composition gives a character, where $\ell \in \mathbb{Z}$ and

$$\langle \chi^m, \lambda^u \rangle = \ell = \sum_i^n a_i b_i$$

where $m = (a_1, \dots, a_n)$ and $u = (b_1, \dots, b_n)$. This number shall come back later

¹Note that if we restricted to group homomorphisms, we would have to restrict the domain $(S^1)^n$ to keep the complete characterization, showing another parallel between the concepts

If $N \cong \mathbb{Z}^n$ is a co lattice, we can canonically associated to it a torus via the (group and regular) isomorphism:

$$N \otimes_{\mathbb{Z}} \mathbb{C}^* \cong (\mathbb{C}^*)^n \quad u \otimes t \mapsto \lambda^u(t) = (t^{u_1}, \dots, t^{u_n})$$

This shows that the geometric structure of the torus is strongly characterized by lattices, allowing us to study tori via these lattices.

Given our basic algebraic object, we would like to now have a family of objects based on this object. This gives the following object:

Definition 1.1.4: Affine Toric Variety

Let V be an irreducible affine variety containing the torus $T_N \cong_{\mathbf{Poly}} (\mathbb{C}^*)^n$ as a Zariski open dense subset such that the action of T_N on itself extends to an algebraic action of T_N on V . Then V is an *affine toric variety*.

Note that an algebraic action $T \times V \rightarrow V$ is a variety morphism. Note that the isomorphism need not also be a group isomorphism, we'll often induce a group structure on T_N via the isomorphism.

Example 1.1: Affine Toric Varieties

1. The simplest such examples are $(\mathbb{C}^*)^n$ and $\mathbb{C}^n$. The action is given by multiplication.
2. The affine n space $\mathbb{A}_{\mathbb{C}}^n$ is an affine toric variety: the natural map $(\mathbb{C}^*)^n \hookrightarrow \mathbb{C}^n$ shows that the torus is an open dense subset, and the action extends trivially.
3. Take the curve $C = V(x^3 - y^2) \subseteq \mathbb{C}^2$, which is the famous curve with a cusp at $(0,0)$. Then this is an affine toric variety with torus

$$C \setminus \{0\} = C \cap (\mathbb{C}^*)^2 = \{(t^2, t^3) \mid t \in \mathbb{C}^*\} \cong \mathbb{C}^*$$

with the variety isomorphism given by $t \mapsto (t^2, t^3)$. Note this is not a normal variety, hence toric varieties need not be normal. The algebraic action given by

$$(t, (x, y)) \mapsto (t^2x, t^3y)$$

and again trivially extends to all of C .

4. The famous twisted surface $V = V(xy - zw) \subseteq \mathbb{C}^4$ is a toric variety with torus:

$$V \cap (\mathbb{C}^*)^4 = \{(t_1, t_2, t_3, t_1 t_2 t_3^{-1}) \mid t_i \in \mathbb{C}^*\} \cong (\mathbb{C}^*)^3$$

Note that this variety is normal. The algebraic action is given by:

$$(t_1, t_2, t_3) \mapsto (t_1x, t_2y, t_3z, t_1 t_2 t_3^{-1}w)$$

5. Take

$$\Phi : \mathbb{C}^2 \rightarrow \mathbb{C}^{d+1} \quad (s, t) \mapsto (s^d, s^{d-1}t, \dots, st^{d-1}, t^d)$$

which can be thought as the embedding of all possible degree d monomials in 2 variables. Let us show the image of φ is a variety. Take coordinates $x_0, \dots, x_d$ for $\mathbb{C}^{d+1}$. Take the ideal $I \subseteq \mathbb{C}[x_0, \dots, x_d]$ generated by 2×2 minors of the matrix:

$$\begin{pmatrix} x_0 & x_1 & \cdots & x_{d-2} & x_{d-1} \\ x_1 & x_2 & \cdots & x_{d-1} & x_d \end{pmatrix}$$

namely, $I = \langle x_i x_{j+1} - x_{i+1} x_j \mid 0 \leq i < j \leq d-1 \rangle$. Then it should be verified that:

$$\Phi(\mathbb{C}^2) = V(I)$$

let $\hat{C}_d = \Phi(\mathbb{C}^2)$. Check that $I(\hat{C}_d) = I$. It shall soon be shown that this variety is irreducible. This variety is often called the *rational normal cone of degree d* , and is a *determinantal variety*. To see that it is a toric ideal,:

$$\Phi((\mathbb{C}^*)^2) = \hat{C}_d \cap (\mathbb{C}^*)^{d+1} \cong (\mathbb{C}^*)^2$$

can you see the isomorphism?

1.1.1 Construction of Affine Toric Variety

There are many ways we can construct Tori. We shall show three ways in this section

From Monomial Maps

Let $T \cong (\mathbb{C}^*)^n$ be a torus with character lattice M , and take a finite subset $S = \{m_1, \dots, m_s\} \subseteq M$. Take the map

$$\Phi_S : T \rightarrow \mathbb{C}^s \quad t \mapsto (\chi^{m_1}(t), \dots, \chi^{m_s}(t))$$

Let $Y_S = \overline{\text{im}(\Phi_S)}$. This is certainly an affine variety. We want to show it is a toric variety. To this end, we need a lemma:

Lemma 1.1.5: Torus Morphism Induces Character Map

Let $\varphi : T_1 \rightarrow T_2$ be a group and regular morphism between tori. Then it induces a map $\hat{\varphi} : M_2 \rightarrow M_1$ where M_i is the character lattice of T_i . If φ is injective, $\hat{\varphi}$ is surjective, and similarly if φ is surjective

Proof :

This is a result straight from dual theory.

Proposition 1.1.6: Image and Closed Subgroup of Torus

1. Let $\varphi : T_1 \rightarrow T_2$ be a group and variety morphism of tori. Then $\text{im } \varphi$ is a torus and closed in T_2
2. If $H \subseteq T$ is a closed irreducible sub-variety of a torus T that is also a subgroup, then H is a torus.

Proof :

1. Let $T' = \text{im } \varphi$. Let us first show that T' is closed. As T_1, T_2 are tori, let ψ_1, ψ_2 be the isomorphism to $(\mathbb{C}^*)^n$ and $(\mathbb{C}^*)^m$ respectively. Consider

$$\psi_2 \circ \varphi \circ \psi_1^{-1} : (\mathbb{C}^*)^n \rightarrow (\mathbb{C}^*)^m$$

which is a toric morphism. It is easy to see that any such map must have a closed image (each image is associated to a lattice, which is canonically associated to a torus). Hence, pushing forward to $\varphi : T_1 \rightarrow T_2$ we see that the image must be closed.

2. As H is a group, it must be linear. As it is closed, it cannot be a discrete subgroup, nor can it be a localization of a discrete subgroup. As $(\mathbb{C}^*)^n$ is irreducible, it can only be a set of dimension $\leq n$. As it is irreducible, it cannot be the union of multiple “faces”, leaving only $H \cong (\mathbb{C}^*)^m$ for some $m \leq n$.

Definition 1.1.7: Lattice Toric Variety

Let T_N be a torus with character lattice M , and let $\mathcal{A} = \{m_1, \dots, m_s\} \subseteq M$ be a set associated to the characters $\chi^{m_i} : T_N \rightarrow \mathbb{C}^*$. Take the map

$$\begin{aligned} \Phi_{\mathcal{A}} : T_N &\rightarrow \mathbb{C}^s \\ \Phi_{\mathcal{A}}(t) &= (\chi^{m_1}(t), \dots, \chi^{m_s}(t)) \in \mathbb{C}^s \end{aligned}$$

Then the affine toric variety $Y_{\mathcal{A}}$ induced by the lattice is defined as the Zariski closure of the image of $\Phi_{\mathcal{A}}$.

Concretely, if we fix a basis for M so that $M \cong \mathbb{Z}^n$, then the s vectors $\mathcal{A} \subseteq \mathbb{Z}^n$ can be seen to form an $n \times s$ matrix with integer entries, and so each such matrix is associated in the right circumstance to a torus. We shall show that all affine toric varieties arise like this. Let us first show that these are indeed affine toric varieties

Proposition 1.1.8: Torus Lattice is Variety

Take a set $S \subseteq M$ and the sublattice $\mathbb{Z}S \subseteq M$. Then Y_S is an affine toric variety whose torus has character lattice $\mathbb{Z}S$, and the dimension of Y_S is given by the rank of $\mathbb{Z}S$.

Proof :

Let $T' = \text{im } \Phi_A$. Then it is closed by the above lemma. Hence, if $\overline{T'}$ is the closure of T' in $\mathbb{C}^s$, then

$$T' = \overline{T'} \cap (\mathbb{C}^*)^s = Y_S \cap (\mathbb{C}^*)^s$$

Hence, $Y_S = \overline{T'}$.

1.1.2 From Toric Ideal

Let us now describe the ideal $I_{\mathcal{A}} \subseteq \mathbb{C}[x_1, \dots, x_n]$ that is associated to toric varieties. If $\Phi_{\mathcal{A}}$ is the toric map from before, then $\Phi_{\mathcal{A}}(f)$ must vanish. To see when it vanishes, notice that if

$$\sum_i a_i m_i = \sum_i b_i m_i$$

then $x^a - x^b$ will vanish: plugging $x = (\chi^{m_1}(t), \dots, \chi^{m_s}(t)) \in \text{im } \Phi_{\mathcal{A}}$, then:

$$\prod_i \chi^{m_i}(t)^{a_i} - \prod_i \chi^{m_i}(t)^{b_i} = \chi^{\sum_i a_i m_i}(t) - \chi^{\sum_i b_i m_i}(t) = 0$$

Thus for every $a, b \in \mathbb{N}^s$ such that $a - b \in \ker \widehat{\Phi}_{\mathcal{A}}$ we have $x^a - x^b \in I(Y_{\mathcal{A}})$. Note that for any $\ell \in \ker \widehat{\Phi}_{\mathcal{A}} \subseteq \mathbb{Z}^s$, we can split it to $\ell = \ell_+ - \ell_-$ where (with the standard basis for $\mathbb{Z}^s$:

$$\ell_+ = \sum_{\ell_i > 0} \ell_i e_i \quad \ell_- = \sum_{\ell_i < 0} \ell_i e_i$$

so then by construction $x^{\ell_+} - x^{\ell_-} \in I(Y_{\mathcal{A}})$. It turns out that the ideal is generated by such elements

Proposition 1.1.9: Toric Ideal

Let $Y_{\mathcal{A}} \subseteq \mathbb{C}^s = \text{Spec}(\mathbb{C}[x_1, \dots, x_s])$ be the affine toric variety induced by $\mathcal{A} \subseteq M$. Then:

$$I(Y_{\mathcal{A}}) = \langle x^{\ell_+} - x^{\ell_-} \mid \ell \in L \rangle = \langle x^{\alpha} - x^{\beta} \mid \alpha, \beta \in \mathbb{N}^s \text{ and } \alpha - \beta \in L \rangle$$

Proof :

David Cox p.15

We give a name to the ideal generated by these elements:

Definition 1.1.10: Toric Ideal

Let $Y_{\mathcal{A}} \subseteq \mathbb{C}^s = \text{Spec}(\mathbb{C}[x_1, \dots, x_s])$ be the affine toric variety induced by $\mathcal{A} \subseteq M$. Then the ideal $I_{\mathcal{A}}$ is called a *toric ideal*. In particular, it is a prime ideal generated by binomials.

Example 1.2: Toric Ideal

1. Take $\mathcal{A} = \{1, 2, 3\}$. Then consider $\widehat{\Phi}_{\mathcal{A}} : \mathbb{Z}^3 \rightarrow \mathbb{Z}$ which will have the kernel generated by:

$$(1, -2, 1) \quad (1, 1 - 1) \quad (2, -1, 0)$$

Hence we get:

$$I(Y_{\mathcal{A}}) = \langle xz - y^2, xy - z, x^2 - y \rangle$$

Notice that $Y_{\mathcal{A}}$ is the twisted cubic.

2. Let $\mathcal{A} = \{2, 3\}$. Show that $Y_{\mathcal{A}}$ is the cuspidal cubic
3. Let $\mathcal{A} = \{(1, 0), (0, 1), (1, 1)\}$. Then

$$I(Y_{\mathcal{A}}) = \langle xy - z \rangle$$

4. Show that $(1, \dots, 1) \in Y_{\mathcal{A}}$ for any $\mathcal{A}$

1.1.3 From Semigroup

We shall now define the embedding without the need to reference the embedding space, that is we define it in an intrinsic way. We shall see that the embeddings shall correspond to choosing generators for an affine semigroup. Recall that a semigroup is a group where invertibility and the identity axiom's are omitted (i.e. it is an associative magma). A semigroup is affine if it is isomorphic to a finitely generated semigroup of $\mathbb{Z}^s$.

Definition 1.1.11: Affine [commutative] Semigroup

Let S be a set. Then $(S, +)$ is an *affine [commutative] semigroup* if $+$ is a binary operation where:

1. $+$ is associative
2. $0 \in S$ is the identity
3. S is finitely generated, so that there is a finite $A \subseteq S$ where $\mathbb{N} \cdot A = S$
4. S embeds in a lattice M .

The identity condition makes it seem like it is more appropriate to say we are working with affine monoid, however the terminology seems to have stuck. To create a variety, we need an algebra:

Definition 1.1.12: Affine Semigroup Algebra

Let M be the character lattice of a torus T . Then the [affine] *semigroup algebra* associated to an affine semigroup $S \subseteq M$ is the $\mathbb{C}$ -algebra:

$$\mathbb{C}[S] = \left\{ \sum_{m \in S} c_m \chi^m \mid c_m \in \mathbb{C}, \text{ finitely many nonzero } c_m \right\}$$

with multiplication induced by the group M :

$$\chi^m \chi^{m'} = \chi^{m+m'}$$

The Laurent polynomial ring $\mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ is given by $\mathbb{C}[\mathbb{Z}^n]$, and in general for $\mathcal{A} = \{m_1, \dots, m_n\}$, we get $\mathbb{C}[\mathbb{N}\mathcal{A}]$.

Proposition 1.1.13: Semigroups and Affine Varieties

Let $S \subseteq M$ be an affine semigroup. Then:

1. $\mathbb{C}[S]$ is an integral domain and a finitely generated $\mathbb{C}$ -algebra
2. $\text{mSpec}(\mathbb{C}[S])$ is an affine toric variety whose dense character lattice is $\mathbb{Z}S$

Proof :

1. Note that $\mathbb{C}[M]$ is isomorphic to the Laurent polynomial making it integral, hence $\mathbb{C}[S]$ is integral, and is finitely generated by definition of the semigroup
2. Pick $\mathcal{A} \subseteq M$ so that $S = \mathbb{N}\mathcal{A}$. Then define $\varphi_{\mathcal{A}}^{\#} : \mathbb{C}[x_1, \dots, x_s] \rightarrow \mathbb{C}[M]$ be the usual pullback $f \mapsto f \circ \Phi_{\mathcal{A}}$. Then we get the short exact sequence:

$$0 \rightarrow I(Y_{\mathcal{A}}) \rightarrow \mathbb{C}[x_1, \dots, x_s] \xrightarrow{\varphi_{\mathcal{A}}^{\#}} \mathbb{C}[S] \rightarrow 0$$

giving

$$\mathbb{C}[S] \cong \frac{\mathbb{C}[x_1, \dots, x_s]}{I(Y_{\mathcal{A}})} \quad \Rightarrow \quad \text{mSpec}(\mathbb{C}[S]) \cong Y_{\mathcal{A}}$$

Then as $\mathbb{Z}\mathcal{A} = \mathbb{Z}S$, the torus of $\text{mSpec}(\mathbb{C}[S])$ has characters $\mathbb{Z}S$, showing density and completing the proof.

1.1.4 Summary

Putting this all together, we get that these are all equivalent:

Theorem 1.1.14: Equivalent Characterization of Toric Variety

Let V be an affine variety. Then the following are equivalent:

1. V Is a toric variety
2. $V \cong Y_{\mathcal{A}}$ for a finite subset $\mathcal{A} \subseteq M$
3. V is given by a toric ideal
4. $V = \text{mSpec}(\mathbb{C}[S])$ for a semigroup S

Proof :

It is simplest to do $2 \Rightarrow 3 \Rightarrow 4 \Rightarrow 2$ (you may not want to write the solution to practice in the future)

1.2 Normal Toric Varieties: Newton Polyhedra

Normal toric varieties come from very simple geometric shapes: they shall come from [rational] cones. To see this, first note that for any lattice $N \cong \mathbb{Z}^n$, we can associate to it $N_{\mathbb{R}} = \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$. Similarly, for a colattice $M = \text{Hom}_{\mathbb{Z}}(N, \mathbb{Z})$, we get $M_{\mathbb{R}} = \text{Hom}_{\mathbb{R}}(N, \mathbb{R}) \cong M \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$.

1.2.1 Cones

Definition 1.2.1: Rational Convex Polyhedral Cone

Let N be a lattice. Then a *Rational Convex Polyhedral Cone* in $N_{\mathbb{R}}$ is a set of the form:

$$\sigma = \text{Cone}(S) = \left\{ \sum_{t \in S} \lambda_n t \mid \lambda_n \in \mathbb{R}_{\geq 0} \right\}$$

where $S \subseteq N$ is finite.

As the cones that are related to toric varieties are all rational, convex, and polyhedral, in this paper we shall just say “cone” understanding that all we are working with cones as given by definition 1.2.1. For any cone $\sigma \subseteq N_{\mathbb{R}}$, the *dual cone* or *cocone* $\sigma^{\vee}$ is the set:

$$\sigma^{\vee} = \{m \in M_{\mathbb{R}} \mid \langle n, m \rangle \geq 0 \text{ } n \in \sigma\}$$

This is a good definition as $(\sigma^{\vee})^{\vee} = \sigma$. Certainly, if σ is rational, $\sigma^{\vee}$ is rational.

Example 1.3: Cones and co-Cones

1. Let $N = \mathbb{Z}^2$. Then there are countably many cones, one for each choice of two vectors in $\mathbb{Z}^2$ which start at different points and have different angles between them. Naturally, you can have more. (Give some examples). (take their dual) **Give the toric variety associated to it.**

2. Do some more examples in higher dimension

Without the finite condition on S , it wouldn't be polyhedral (flat object, in a geometric sense, like intersections of half-planes), for ex. you can have a “rounded” cone if S isn't finite. As $S \subseteq N$, it is rational. As $\lambda_n \geq 0$, it is a cone.

Cone have a natural notion of dimension: for $\sigma \subseteq N_{\mathbb{R}}$, $\dim \sigma = \min \{ \dim V \mid \sigma \subseteq V \subseteq N_{\mathbb{R}}, V \text{ vector subspace} \}$, that is the dimension of σ is the dimension of the smallest linear subspace containing it. Next, there is a natural generalization of the notion of face from the classical notion of polyhedron. For some of the following definitions, we shall require considering the following sets: $H_m = \{n \in N_{\mathbb{R}} \mid \langle n, m \rangle = 0\}$ and $H_m^+ = \{n \in N_{\mathbb{R}} \mid \langle n, m \rangle \geq 0\}$.

Definition 1.2.2: Face of Cone

Let σ be a cone, and pick some $m \in M_{\mathbb{R}}$. Then a *face* of σ is:

$$\tau = \sigma \cap H_m$$

If $\dim \tau = 1$, we call τ a *ray*.

Faces of [rational] cones are [rational] cones. As $0 \in \sigma^\vee$ (the “generic” point), we see that σ can itself be considered a face. We thus get a partial ordering $\tau \preceq \sigma$. If $\tau \preceq \sigma$ where $\dim \sigma = n$, then if $\dim \tau = k$, $\text{codim} \tau^\vee \leq k$ (we shall see when equality happens soon)

Example 1.4: Faces

Take some of the examples above and find some faces

Proposition 1.2.3: Properties of Faces

Let σ be a cone. Then:

1. $\{m_i\}_{i=1}^s \subseteq M$ generate $\sigma^\vee$ if and only if $\sigma = H_{m_1}^+ \cap \cdots \cap H_{m_s}^+$
2. If $\tau \preceq \sigma$ and $\tau' \preceq \sigma$, then $\tau \cap \tau' \preceq \sigma$
3. If $\tau' \preceq \tau$ and $\tau \preceq \sigma$, then $\tau' \preceq \sigma$
4. If $\tau \preceq \sigma$, $v, w \in \sigma$ such that $v + w \in \tau$, then $v, w \in \tau$.
5. Faces of σ are in inclusion-reversing bijection to the faces of $\sigma^\vee$

Proof :
exercise

We will require three more concepts that shall naturally come up:

Definition 1.2.4: Strongly Convex Cones

Let $\sigma \subseteq N_{\mathbb{R}}$ be a cone. Then σ is called *pointed* or *strongly convex* if

$$\sigma \cap (-\sigma) = \{0\}$$

Show that this is equivalent to saying $\{0\} \preceq \sigma$ and $\dim \sigma^{\vee} = n$. If $\rho \preceq \sigma$ is a ray and σ is strongly convex, then there is a unique point $\nu_p \in N$ such that

$$\rho \cap N = \mathbb{N}\nu_p$$

Definition 1.2.5: Smooth Cone, Simplicial Cones

Let $\sigma \subseteq N_{\mathbb{R}}$ be a strongly convex cone. Then:

1. σ is called *smooth* if a set of minimal generators is a subset of a $\mathbb{Z}$ -basis for $\mathbb{R}$.
2. σ is called *Simplicial* if a set of minimal generators is linearly independent over $\mathbb{R}$.

Example 1.5: Different Types of Cones

HERE

let us now see how we can associate to a cone an affine toric variety. Let:

$$S_{\sigma} = \sigma^{\vee} \cap M$$

Then:

Proposition 1.2.6: Gordan's Lemma

Let S_{σ} be the semigroup defined above. Then S_{σ} is affine.
As a consequence, $\text{mSpec}(\mathbb{C}[S_{\sigma}])$ is an affine toric variety

Proof :

(do this exercise)

Proposition 1.2.7: Dimension of Normal Pointed Cone

Let $\text{mSpec}(\mathbb{C}[S_{\sigma}])$ be an affine toric variety corresponding to $\sigma \subseteq N_{\mathbb{R}}$. Then

$$\dim(\text{mSpec}(\mathbb{C}[S_{\sigma}])) = n \text{ if and only if } \sigma \text{ is pointed}$$

In this case, $\text{mSpec}(\mathbb{C}[S_{\sigma}])$ has character lattice $M = \text{Hom}_{\mathbb{Z}}(N, \mathbb{Z})$.

Proof :

by proposition 1.1.8,

$$(\mathrm{mSpec}(\mathbb{C}[S_\sigma]) = \mathrm{rank}(\mathbb{Z}S_\sigma)$$

Then we get that

$$\mathbb{Z}S_\sigma = M \quad \text{if and only if} \quad \sigma \text{ is strongly convex}$$

completing the proof.

Example 1.6: Toric Varieties and Cones

1. Take $\sigma = \mathrm{Cone}(e_1, \dots, e_r) \subseteq \mathbb{R}^n = (\mathbb{Z}^n)_{\mathbb{R}}$ with $r \leq n$ with the standard basis. Then σ is pointed, and

$$\sigma^\vee = \mathrm{conv}(\delta_1, \dots, \delta_r, \delta_{r+1}^\pm, \dots, \delta_n^\pm)$$

Hence the associated toric variety is:

$$\mathrm{mSpec}(\mathbb{C}[x_1, \dots, x_r, x_{r+1}^\pm, \dots, x_n^\pm]) \cong \mathbb{C}^r \times (\mathbb{C}^*)^{n-r}$$

Note that changing the basis of $\mathbb{Z}^n$, the corresponding toric variety is the produce of affine space and tori.

2. (ToricNotes p.15)

Let us next find a natural generating set for the affine semigroup

Proposition 1.2.8: Hilbert Basis For Semigroup

re-write this statement, not my style Let $H = \{m \in S_\sigma \mid m \text{ is irreducible}\}$. Then:

1. H is finite and generates S_σ
2. H contains a minimal generating set
3. H is the (a?) minimal set of generators ordered by inclusion

The set H is called the *Hilbert basis* of S_σ (or $\sigma^\vee$)

Proof :

(12, prop 1.2.23)

1.2.2 Characterizing Points

move this section? Make this flow in your way

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